

Sentinel-Bio+: Students Powering the Next Era of Human Spaceflight

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Introduction

The San Diego Space Institute (SDSI), together with San José State University (SJSU) and San Diego State University (SDSU), is training the next generation of aerospace professionals through Sentinel-Bio+, a hands-on CubeSat program directly tied to NASA's human spaceflight and heliophysics priorities.

Two Proposals, One Mission

SJSU Proposal – Bus Development

At San José State University, Professor Periklis Papadopoulos (PI) leads the spacecraft bus design effort, building on SJSU's TechEdSat heritage with NASA Ames. Co-investigator Sean Casey (SDSI) provides nonprofit coordination and mission-level oversight. Students follow NASA's Systems Engineering Handbook, progressing through SRR, PDR, and CDR, before delivering flight-ready bus subsystems.

SDSU Proposal – Payload Integration

At San Diego State University, Professor Ahmad Bani Younes (PI, SDSU Space Lab) guides students in developing a compact payload combining a radiation detector, MEMS accelerometer, and fluxgate magnetometers. He is joined by Sean Casey (SDSI, Co-I) and Industry Mentor Dave Strobel (former CEO, Space Micro), ensuring strong connections to both NASA and the commercial aerospace sector.

Why It Matters

Radiation and drag are the two invisible forces that define risk in low Earth orbit. For astronauts aboard Artemis missions or private space stations, understanding these hazards is critical. For satellites, drag-driven orbital decay is a billion-dollar challenge. By uniting student-built buses with student-integrated payloads under SDSI mentorship, Sentinel-Bio+ creates both new science and new scientists.

The Workforce Impact

Through Sentinel-Bio+:

- Students gain NASA-style systems engineering experience.
- Teams conduct design reviews, prototype building, and environmental testing.
- Alumni pathways extend into NASA, SpaceX, and leading aerospace startups.
- CaSGC's \$15k-\$25k investment per project is amplified by SDSI donor matching, yielding high educational returns.

Student Perspective: Why It Matters to Them

For many participants, Sentinel-Bio+ is their first opportunity to contribute to a real space mission. One student reflected that before joining the project, they had never worked on anything space-related, but hearing about the plan to create a constellation of satellites sparked immediate excitement. They described the experience as a chance to finally gain hands-on skills with satellite systems, expand their technical abilities, and connect classroom knowledge to a future aerospace career. Above all, they expressed gratitude for the opportunity to be part of a space mission and to build something that could directly impact human spaceflight.

Next Steps

With CaSGC seed funding, Sentinel-Bio+ begins training and prototype work in late 2025. SDSI continues raising donor support to fund hardware, launch, and operations for a six-satellite constellation in 2027.